Separable Differential Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

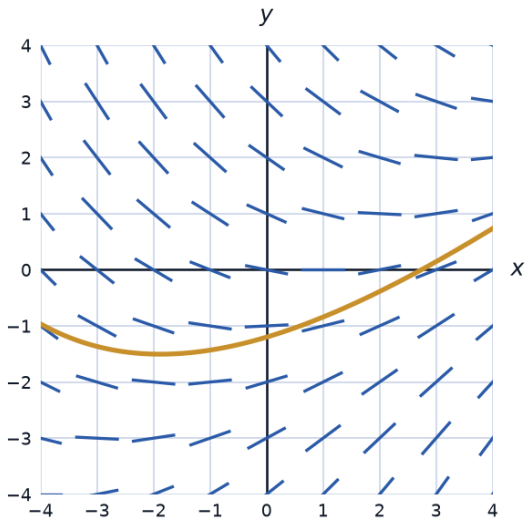
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving by separation of variables**

Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

2 **[Foundational]** • **Target: Growth, decay, and applications**

A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.



a Solve for $P(t)$.

b Find $P(20)$, to the nearest whole number.

3 **[Foundational]** • **Target: Growth, decay, and applications**

Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a $20°$C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.

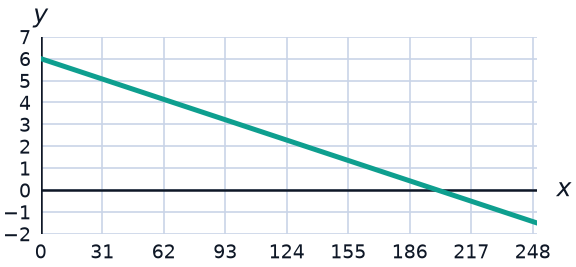
4 **[Foundational]** • **Target: More separable equations**

Solve $\frac{dy}{dx} = y(1 - y)$ (logistic-type equation), given $y(0) = \frac{1}{2}$, using partial fractions $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Mixing and applied models

A tank has 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution drains at the same rate, so the volume stays constant. Let $y(t)$ be the amount of salt (kg) at time t. The differential equation is $\frac{dy}{dt} = 6 - \frac{3y}{100}$.

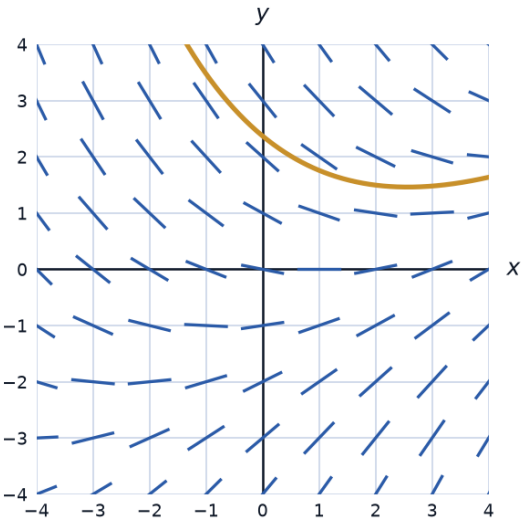


a Find the equilibrium value of y (where $\frac{dy}{dt} = 0$).

b Explain, in words, what this equilibrium represents physically.

6 [Foundational] • Target: Slope fields and qualitative behavior

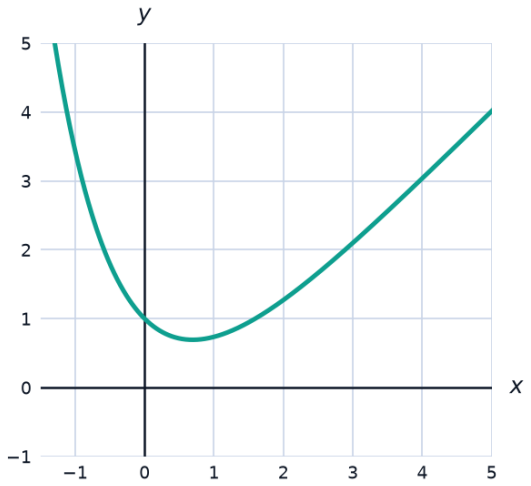
For the differential equation $\frac{dy}{dx} = x - y$, find the slope of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$.



© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Slope fields and qualitative behavior

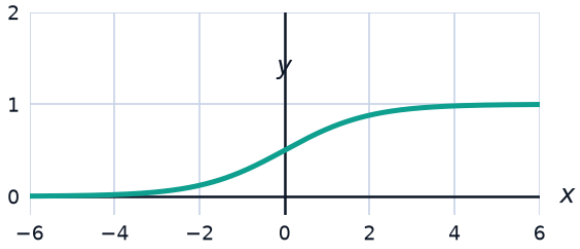
Verify that $y = x - 1 + 2e^{-x}$ is a solution to $\frac{dy}{dx} = x - y$, by substituting into both sides.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Visualizing a logistic solution curve

The graph shows the logistic solution $y = \frac{e^x}{1+e^x}$ found earlier for $\frac{dy}{dx} = y(1-y)$, $y(0) = \frac{1}{2}$. Use the graph to describe the long-run behavior of y as $x \to \infty$ and as $x \to -\infty$.



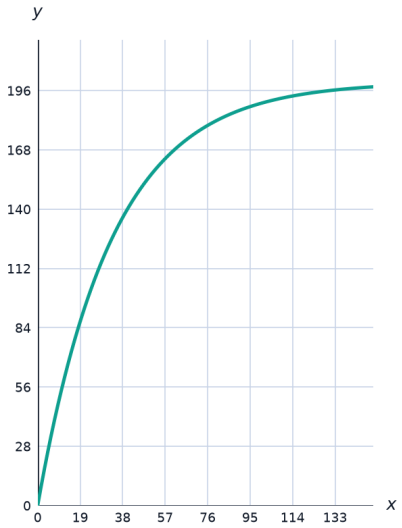
9 [Intermediate] • Target: Another applied separable equation

Radioactive decay follows $\frac{dA}{dt} = -kA$. If a sample decays from 80 g to 20 g in 30 days, find k and the half-life of the substance.

© ClipSAT — clipsat.org | All Rights Reserved

10 **[Intermediate]** • **Target: Verifying a solution to an applied model**

For the mixing problem $\frac{dy}{dt} = 6 - \frac{3y}{100}$ with $y(0) = 0$, verify that $y(t) = 200(1 - e^{-0.03t})$ is a solution, and confirm it approaches the equilibrium 200 found earlier as $t \to \infty$.

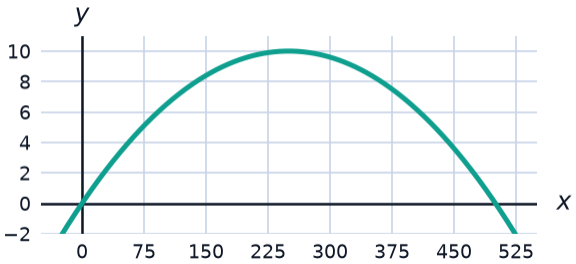


11 **[Intermediate]** • **Target: Verifying a solution to an applied model**

Solve $\frac{dy}{dx} = e^{x-y}$, given $y(0) = 0$, using the fact that $e^{x-y} = e^x \cdot e^{-y}$.

12 **[Intermediate]** • **Target: Verifying a solution to an applied model**

A pond's fish population P grows according to $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{500}\right)$, a logistic model with carrying capacity 500. Without solving the equation explicitly, determine the equilibrium population levels (where $\frac{dP}{dt} = 0$), and state which one is realistic for a growing population starting from $P(0) = 50$.



13 **[Intermediate]** • **Target: Verifying a solution to an applied model**

Solve $\frac{dy}{dx} = \frac{\cos x}{y}$, given $y(0) = -2$, and state why the negative root must be chosen.

14 **[Intermediate]** • **Target: A final applied separable model**

A drug is eliminated from the bloodstream according to $\frac{dC}{dt} = -0.2C$, where C is concentration (mg/L) and t is hours. If the initial dose gives $C(0) = 50$ mg/L, find $C(t)$, then find how many hours it takes for the concentration to drop below 5 mg/L, to one decimal place.

© ClipSAT — clipsat.org | All Rights Reserved

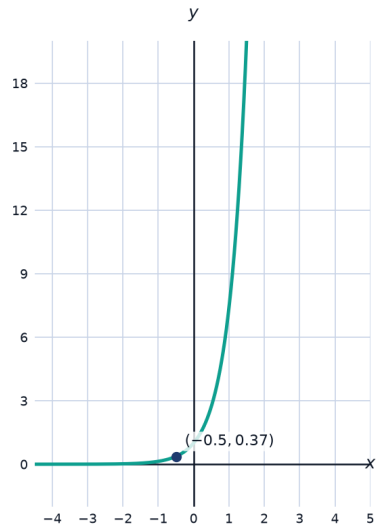
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Slope fields and qualitative behavior • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Level 2 · MediumFind the particular solution to the differential equation $\frac{dy}{dx} = \frac{x}{y}$ with initial condition $y(0) = -4$. State the domain of the solution.

16 [Challenging] • Target: Visualizing a logistic solution curve • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Level 1 · FoundationsWhich of the following functions is a solution to the differential equation $y'' + 4y = 0$? (A) $y = e^{2x}$ (B) $y = \sin(2x) - 3\cos(2x)$ (C) $y = 4\cos(4x)$ (D) $y = x^2 - 4$



17 [Challenging] • Target: Another applied separable equation • Cross-Lesson: 6.01 Related Rates

Level 3 · Advanced / AP FRQLet $y = f(x)$ be the particular solution to $\frac{dy}{dx} = xy^2$ with $f(1) = -1$. Use Euler's method with step size $\Delta x = 0.5$ starting at $x = 1$ to approximate $f(2)$, and determine whether the approximation is an overestimate or underestimate by computing $\frac{d^2y}{dx^2}$ at $(1, -1)$.

18 [Challenging] • Target: Separable equations with algebraic solutions • Cross-Lesson: 6.02 Optimization and Curve Sketching

Solve the separable DE $\frac{dy}{dx} = \frac{x+1}{y}$ given $y = 2$ when $x = 0$.

19 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 7.01 Integration by Substitution

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

20 [Challenging] • Target: A final applied separable model • Cross-Lesson: 7.02 Integration by Parts

Solve the equation $z^3 + 8 = 0$, expressing all solutions in Cartesian form $a + bi$.

© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving by separation of variables

A population grows according to the logistic model $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{315}\right)$. Find the population P at the point of maximum growth.

a. 630

b. 78.75

c. 315

d. 157.5

22 [Foundational] • Target: Growth, decay, and applications

Consider the differential equation $\frac{dy}{dx} = 2x$. Which of the following is a particular solution passing through the point $(1, 3)$?

a. $y = x^2 + 2$

b. $y = x^2 + 1$

c. $y = 2x + 1$

d. $y = 2x^2 - 1$

23 [Foundational] • Target: More separable equations

Exponential decay: $\frac{dN}{dt} = -kN$, $k > 0$. Solution:

a. $N = N_0 e^{kt}$

b. $N = N_0/(1 + kt)$

c. $N = N_0 - kt$

d. $N = N_0 e^{-kt}$

24 [Foundational] • Target: Mixing and applied models

Separable ODE $\frac{dy}{dx} = xy$. Solution:

a. $y = x^2/2 + C$

b. $y = Ce^x$

c. $y = Ce^{x^2/2}$

d. $y = Ce^{x^2}$

25 [Foundational] • Target: Slope fields and qualitative behavior

Separable ODE: $\frac{dy}{dx} = \frac{x}{y}$. Solution (implicit):

a. $x^2 + y^2 = C$

b. $y^2 = x^2$

c. $y = x + C$

d. $y^2 = x^2 + C$

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Visualizing a logistic solution curve

Logistic equation steady states (equilibria):

a. $P = 0$ only

b. $P = K$ only

c. $P = 0$ and $P = K$

d. $P = K/2$

27 [Foundational] • Target: Another applied separable equation

The slope field for a differential equation is shown above. The differential equation is of the form $\frac{dy}{dx} = ky$ for some constant k. If the solution curve passes through the point $(0, 1)$ and $(1, 2)$, find the value of k.

a. $\ln 2$

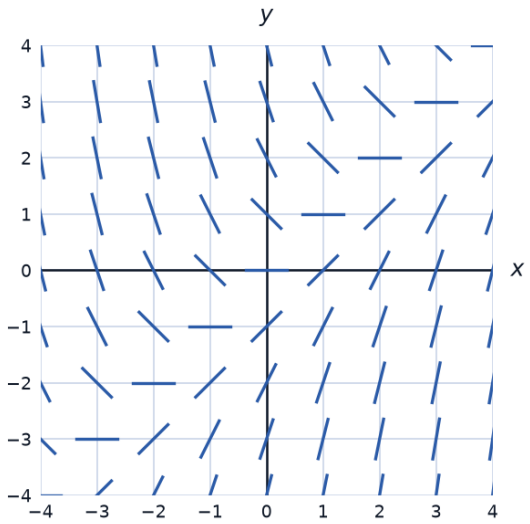
b. 2

c. $\frac{1}{2}$

d. 1

28 [Foundational] • Target: Separable equations with algebraic solutions

Consider the slope field shown below. Which of the following differential equations could represent this slope field, assuming the slopes are defined at all points except where division by zero occurs? $\frac{dy}{dx} = \frac{x}{y}$



a. $\frac{dy}{dx} = \frac{x}{y}$

b. $\frac{dy}{dx} = \frac{y}{x}$

c. $\frac{dy}{dx} = x - y$

d. $\frac{dy}{dx} = x + y$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 **[Intermediate]** • **Target: Verifying a solution to an applied model**

A population of bacteria follows a logistic growth model where $P(t)$ is the population at time t (in hours). The differential equation governing the population growth is $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{1000}\right)$. If the initial population is $P(0) = 200$, use Euler's method with a step size of $\Delta t = 1$ hour to approximate the population after 1 hour, $P(1)$.

a. 210

b. 208

c. 212

d. 209

30 **[Intermediate]** • **Target: A final applied separable model**

Identify the general solution to the first-order differential equation $\frac{dy}{dx} = 3y$.

a. $y = C\ln(3x)$

b. $y = \frac{3}{2}x^2 + C$

c. $y = Ce^{-3x}$

d. $y = Ce^{3x}$

31 **[Intermediate]** • **Target: Solving by separation of variables**

For the differential equation $\frac{dy}{dx} = 1x - 5y$, find the slope of the solution curve at the point $(3, 2)$.

a. -9

b. -5

c. 7

d. -7

32 **[Intermediate]** • **Target: Growth, decay, and applications**

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1e^{r_1x} + C_2e^{-r_1x}$

b. $y = C_1e^{r_1x}$

c. $y = (C_1 + C_2)e^{2r_1x}$

d. $y = (C_1 + C_2x)e^{r_1x}$

33 **[Intermediate]** • **Target: More separable equations**

For the autonomous equation $y' = -y^2 - y$, which equilibrium is unstable?

a. $y = -1$

b. $y = 0$

c. both are stable

d. neither is stable

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Mixing and applied models

For the autonomous equation $y' = -y^2 + 5y$, which equilibrium is stable?

a. $y = 5$

b. $y = 0$

c. both are stable

d. neither is stable

35 [Intermediate] • Target: Slope fields and qualitative behavior

Bernoulli equation $y' + P(x)y = Q(x)y^n$: substitution to linearise is:

a. $v = y^n$

b. $v = y^{1-n}$

c. $v = y^2$

d. $v = 1/y$

36 [Intermediate] • Target: Visualizing a logistic solution curve

Homogeneous 2nd-order: characteristic equation $r^2 + r - 6 = 0$. Roots:

a. $r = 2, -3$

b. $r = -2, 3$

c. $r = 1, -6$

d. $r = 3, 2$

37 [Intermediate] • Target: Another applied separable equation

Second-order constant-coefficient ODE $y'' - y = 0$: characteristic equation:

a. $r - 1 = 0$

b. $r^2 + 1 = 0$

c. $r^2 - 1 = 0$

d. $r^2 + r - 1 = 0$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Separable equations with algebraic solutions • Cross-Lesson: 6.02 Optimization and Curve Sketching

Radioactive decay: A has a 10-year half-life. After 30 years, fraction remaining:

a. 1/4

b. 1/8

c. 1/6

d. 1/3

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 7.01 Integration by Substitution

Linear homogeneous 2nd-order ODE: if y_1, y_2 are solutions, so is:

a. y_1^2

b. $y_1 y_2$

c. y_1'

d. $y_1 + y_2$

40 [Challenging] • Target: A final applied separable model • Cross-Lesson: 7.02 Integration by Parts

First-order linear ODE solution structure: general solution = homogeneous solution +

a. particular solution

b. another homogeneous solution

c. the initial condition

d. zero

41 [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 8.01 Area Between Curves

General solution of $\frac{dy}{dx} = 2x$:

a. $y = x^2$

b. $y = 2$

c. $y = x^2 + C$

d. $y = 2x + C$

42 [Challenging] • Target: Growth, decay, and applications • Cross-Lesson: 8.02 Volumes of Revolution

Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^x$

c. $y = 2x + 2$

d. $y = xe^x$

43 [Challenging] • Target: More separable equations • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Method of undetermined coefficients: for $y'' + y = e^x$, try particular solution:

a. $y_p = Ae^x$

b. $y_p = Ae^{2x}$

c. $y_p = Axe^x$

d. $y_p = A\sin x$

44 [Challenging] • Target: Mixing and applied models • Cross-Lesson: 1.01 Precalculus Review for Calculus

Resonance in $x'' + \omega_0^2 x = F_0 \cos(\omega_0 t)$: particular solution grows as:

a. $\sin(\omega_0 t)$

b. $e^{\omega_0 t}$

c. $t^2 \cos(\omega_0 t)$

d. $t\cos(\omega_0 t)$

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Slope fields and qualitative behavior • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Cauchy-Euler $x^2y'' + xy' + y = 0$: characteristic roots $r = \pm i$. Solution:

a. $y = C_1x^i + C_2x^{-i}$

b. $y = C_1\cos(\ln x) + C_2\sin(\ln x)$

c. $y = C_1\ln x + C_2$

d. $y = x(C_1\cos x + C_2\sin x)$

© ClipSAT — clipsat.org | All Rights Reserved